Northwestern 2 at BAREC Shared Task 2026: CAMeLBERT-Mix for Fine-Grained Arabic Readability Assessment

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Abstract

This paper presents the Northwestern University in Qatar (NU-Q) submission to Task 1 of the BAREC Shared Task 2026 on fine-grained Arabic readability assessment. The task requires predicting the readability level of Arabic sentences across 19 ordered readability levels. Our system fine-tunes CAMeLBERT-Mix as a 19-class classifier using the official BAREC training data and cross-entropy optimization. We evaluate the proposed system using Quadratic Weighted Kappa (QWK), the official shared-task evaluation metric, together with classification accuracy. Our submitted system achieves a validation QWK of 0.7393 and an accuracy of 45.96%, indicating that pretrained Arabic language models provide a strong baseline for fine-grained Arabic readability assessment.

1 Introduction

Automatic readability assessment aims to estimate the reading difficulty of text and plays an important role in educational technology, language learning, and intelligent tutoring systems. Recent advances in pretrained language models have substantially improved readability prediction by learning contextual representations for morphologically rich languages such as Arabic.

The BAREC Shared Task 2026 addresses fine-grained Arabic readability assessment by requiring systems to assign each Arabic sentence to one of 19 ordered readability levels. Unlike coarse-grained readability prediction, distinguishing between neighboring levels is particularly challenging because they often differ only in subtle lexical, syntactic, and semantic characteristics. The ordinal nature of the labels further increases the difficulty, as prediction errors frequently occur between adjacent readability levels.

In this paper, we present the Northwestern University in Qatar (NU-Q) submission to Task 1 of the BAREC Shared Task 2026. Our approach fine-tunes CAMeLBERT-Mix as a 19-class classifier using only the official BAREC training data under the Strict Track setting. The system adopts a simple yet effective fine-tuning strategy based on cross-entropy optimization, allowing pretrained contextual representations to capture linguistic cues associated with different readability levels. Our submitted system achieves a validation QWK of 0.7393 and an accuracy of 45.96%.

2 Background

2.1 Shared Task Description

The BAREC Shared Task 2026 Task 1 Elmadani et al. (2026) focuses on fine-grained Arabic readability assessment, where each input consists of a single Arabic sentence and the system predicts one of 19 ordered readability levels. We participated in the Strict Track, which restricts systems to using only the official shared-task training data together with publicly available pretrained language models. The official evaluation metric is Quadratic Weighted Kappa (QWK), which accounts for the ordinal relationship between neighboring readability levels. The task builds upon the BAREC benchmark and the annotation guidelines proposed for fine-grained Arabic readability assessment Habash et al. (2025),Elmadani et al. (2025b),Elmadani et al. (2025a).

2.2 Related Work

Earlier Arabic readability research introduced both linguistic metrics and educational resources. El-Haj and Rayson (2016) introduced OSMAN, an Arabic readability metric and tool that supports text with and without diacritics, adapting conventional readability formulas through Arabic syllable counts and the Faseeh factor. Alhafni et al. (2024) introduced the SAMER Corpus, a manually annotated Arabic parallel corpus for text simplifi-

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cation targeting school-aged learners, with word- and document-level readability annotations and two simplified versions of each text. El-Haj et al. (2024) introduced DARES, a dataset of Saudi school materials for coarse- and fine-grained readability assessment, and evaluated five Arabic transformer models, with CAMeLBERT-Mix performing best across all input settings.

Beyond these resources, previous research has explored Arabic readability assessment at different levels of granularity. Liberato et al. (2024) compared rule-based and pretrained language model approaches to Arabic readability assessment at the word and sentence-fragment levels, finding that combining techniques achieved the best performance. Hazim et al. (2022) presented a Google Docs add-on for Arabic word-level readability visualization and assisted text simplification, supporting the identification and simplification of difficult words.

Recent efforts have significantly advanced Arabic readability assessment through the development of standardized annotation resources. Habash et al. (2025) introduced detailed annotation guidelines for fine-grained sentence-level Arabic readability, defining 19 ordered readability levels and establishing a consistent annotation framework based on linguistic complexity and reader proficiency.

Building on these guidelines, Elmadani et al. (2025b) introduced the Balanced Arabic Readability Evaluation Corpus (BAREC), a large-scale, fine-grained corpus for Arabic readability assessment. BAREC consists of 69,441 manually annotated sentences spanning more than one million words across 19 readability levels.

To further advance research in Arabic readability assessment, Elmadani et al. (2025a) organized the BAREC Shared Task 2025, which established a standardized benchmark for sentence- and document-level readability assessment under multiple resource settings.

More directly related to our work, Ibrahim et al. (2025) proposed a two-stage Strict-Track approach for BAREC 2025, fine-tuning BAREC-tuned AraBERT checkpoints with Weighted Kappa Loss to align training with QWK. Our submission instead investigates CAMeLBERT-Mix for the 19-class sentence-level task using cross-entropy optimization.

3 System Overview

Our system formulates fine-grained Arabic readability assessment as a 19-class sentence classification task. We fine-tune CAMeLBERT-Mix using the official BAREC 2026 training data under the Strict Track setting. The system consists of sentence tokenization, contextual encoding with CAMeLBERT-Mix, and prediction through a linear classification head.

One of the main challenges is distinguishing between neighboring readability levels. Given an input sentence x, the model predicts the probability of each readability level using the softmax function:

$$P(y = i \mid x) = \frac{\exp(z_i)}{\sum_{j=1}^{19} \exp(z_j)}, \tag{1}$$

The model is optimized using the cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{19} y_i \log P(y = i \mid x), \tag{2}$$

where z_i is the logit for class i, and y_i is the one-hot ground-truth indicator.

3.1 Model

We formulate Arabic sentence-level readability assessment as a 19-class classification task.Our system uses CAMeL-Lab/bert-base-arabic-camelbert-mix, a pretrained Arabic transformer model from the CAMeLBERT family Inoue et al. (2021). A sequence classification head with 19 output neurons is added, and the entire model is fine-tuned on the official BAREC 2026 training data to predict the readability level of each input sentence.

The official BAREC 2026 Shared Task dataset is used for training and validation. Since the original readability labels range from 1 to 19, they are mapped to integer indices from 0 to 18 before training. Input sentences are tokenized using the CAMeLBERT-Mix fast tokenizer and truncated to a maximum sequence length of 128 tokens. Dynamic padding is applied at the batch level using `DataCollatorWithPadding`, enabling efficient GPU utilization during training.

3.2 Training Strategy

The model is fine-tuned using the Hugging Face `Trainer` framework for three epochs. We use a learning rate of 2×10^{-5}, a weight decay of 0.01, a training batch size of 16, and a validation batch

size of 32. Mixed-precision (FP16) training is enabled, and all experiments are conducted with a fixed random seed of 42.

Following the official evaluation protocol, model selection is based on the Quadratic Weighted Kappa (QWK) score computed on the validation set. Evaluation and checkpoint saving are performed after every epoch, and the checkpoint achieving the highest validation QWK score is automatically restored. In addition to QWK, classification accuracy is reported as a complementary metric.

4 Experimental Setup

4.1 Dataset

We use the official BAREC 2026 Shared Task dataset (CAMeL-Lab/BAREC-Shared-Task-2026-sent) under the Strict Track setting. The dataset contains 54,845 training sentences and 7,310 validation sentences annotated with 19 ordered readability levels. Original labels are mapped from the range 1–19 to 0–18 before training. Input sentences are tokenized using the CAMeLBERT-Mix fast tokenizer with a maximum sequence length of 128 tokens. Dynamic padding is applied at the batch level using `DataCollatorWithPadding`.

4.2 Training Configuration

The model is fine-tuned using the Hugging Face `Trainer` framework for three epochs with a learning rate of 2×10^{-5}, a weight decay of 0.01, a training batch size of 16, and a validation batch size of 32. Mixed-precision (FP16) training is enabled, and all experiments are conducted with a fixed random seed of 42.

Experiments were conducted using Transformers 4.55.4, Hugging Face Hub 0.34.4, Datasets 4.0.0, and Accelerate 1.10.1 on an NVIDIA Tesla T4 GPU with CUDA support.

The results demonstrate that CAMeLBERT-Mix provides a competitive baseline for fine-grained Arabic readability assessment. The achieved QWK score indicates that pretrained contextual representations effectively capture lexical and syntactic cues associated with different readability levels.

Despite the encouraging performance, readability assessment remains challenging due to the subtle differences between neighboring readability levels. Since the task involves ordinal labels, many prediction errors are expected to occur between adjacent classes rather than distant ones. Future work may benefit from incorporating ordinal-aware loss functions, ensemble models, or additional linguistic features to further improve performance.

Table 1: Performance on the official BAREC 2026 validation set.

Metric	Score
Quadratic Weighted Kappa (QWK)	0.7393
Accuracy	45.96%
Validation Loss	1.5518

4.3 Evaluation Metrics

Following the official shared-task protocol, Quadratic Weighted Kappa (QWK) is used as the primary evaluation metric. The checkpoint with the highest validation QWK score is selected for the final evaluation. Accuracy is additionally reported as a complementary metric.

5 Results

Table 1 reports the performance of the proposed CAMeLBERT-Mix model on the official BAREC 2026 validation set. Quadratic Weighted Kappa (QWK) is used as the primary evaluation metric. The model achieves a QWK score of **0.7393**, an accuracy of **45.96%**, and a validation loss of **1.5518**.

The difference between QWK and exact classification accuracy reflects the ordinal nature of the task, where predictions close to the gold readability level are penalized less severely than distant predictions. This suggests that exact discrimination among neighboring readability levels remains challenging.

6 Conclusion

This paper presented the Northwestern University in Qatar (NU-Q) submission to Task 1 of the BAREC 2026 Shared Task on fine-grained Arabic readability assessment. We formulated the task as a 19-class sentence classification problem and fine-tuned CAMeLBERT-Mix using only the official training data under the Strict Track setting. The experimental results show that pretrained Arabic language models provide a strong baseline for fine-grained readability assessment. The proposed system achieves a validation QWK score of **0.7393**, demonstrating the effectiveness of pretrained Arabic language models for fine-grained readability assessment. Future work will investigate ordinal-aware learning objectives, data augmentation strate-

gies, and ensemble methods to further improve performance.

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